

THE GLUON AS CURVATURE ARTIFACT: STRONG BINDING FROM THE INNER-OUTER CASIMIR ON THE OCTONION CARRIER

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ABSTRACT. Position in the publication chain. This is a Tier-2 closure note in the HQIV preprint series. It unites three preceding strands: the dynamic inner-outer Casimir realisation of the mass spectrum and its curvature primitive on the T12/T13 Hopf carrier [2], the sector-by-sector projection of the discrete octonion action onto the Standard-Model Lagrangian (including the strong kinetic aggregate) [1], and the composite-trace derivation of nucleon masses and binding energies from the same 8×8 network at lock-in [3]. The present note makes explicit what the architecture already exhibits: there are no fundamental gluon fields. The entire strong-binding phenomenology (hadron mass deficits, nuclear binding, heavy stabilisation) is the geometric inner-Casimir trapping factor on the strong octonion channels, identical in origin, functional form, and scale to the binding mechanism that generates the lepton spectrum and the dynamic electroweak vev.

Result. The force-sector map assigns the four octonion-extension channels $a \in \{4, 5, 6, 7\}$ to the strong sector; the corresponding kinetic term is the direct projection of the single abelian octonion kinetic density L_O^{kin} . Binding at every scale—hadronic, nuclear, and the heavy stabilisation gap—is realised by one and the same 8×8 network sum $E_{\text{bind}} = \sum_k w_k \cdot \alpha_{\text{eff}}(m)$ (or its continuous- ξ lift) on the $\mathfrak{so}(8)$ generators. The “QCD” or “gluonic” weights are simply the projection of that network onto the strong channels. The shell coupling α_{eff} is driven by the curvature primitive whose normalisation is the $\omega K(\xi)$ factor of the inner-outer Casimir balance on the T12/T13 carrier. The Lean comment labelling the inner trapped-Casimir factor “(trapped, binding, heavy/gluonic)” is therefore the literal statement of the framework: the gluon is the continuum name for the curvature artifact generated when the strong-channel projection of the carrier is viewed through a non-geometric lens. The uniformity of the network formula across all binding hierarchies is the central mathematical content of the bound-states module.

Claim discipline. Proved: the sector projection (abelian only), the uniform network-binding spine, the shared curvature primitive $\omega K(\xi)$ and its discrete sampling $\alpha_{\text{eff}}(m)$, and the explicit re-use of the same inner-trapping functional for both the heavy lepton gap and the “gluonic” stabilisation. Not claimed: a dynamical non-abelian $\text{SU}(3)$ gauge theory, a gluon condensate order parameter, or a derivation of confinement as a topological transition. The optional $\mathfrak{su}(3)$ chart certificate supplies only the structure constants on a finite representation; it does not add propagating gluon degrees of freedom.

Patch-ontology reader contract. The discrete octonion channels, the Hopf-shell contact data (T12/T13), the composite-trace weights on the 28 generators of $\mathfrak{so}(8)$, and the curvature primitive evaluated on the temperature ladder are the ontology. Continuum language that speaks of “gluons”, “gluon condensates”, or “color flux tubes” is a calculational shorthand for the geometric binding already present on the strong-preferring directions of the carrier. It does not introduce new fundamental fields.

Patch QFT and discrete numerics (reader contract). HQIV (Horizon-Quantized Informational Vacuum) is formulated as a *patch theory*: quantum fields, actions, and physical readouts are attached to discrete patches—shell indices $m \in \mathbb{N}$, finite spacetime index types (e.g. Fin 4), and horizon-limited charts—rather than to a hypothetical fundamental smooth spacetime obtained as a mesh-refinement or ultraviolet limit. Accordingly, when this text refers to “QFT,” it means *patch QFT*: patching rules, finite measurement ledgers, and variational identities on the discrete spine; it is not the claim that the theory is *defined* by recovering ordinary continuum quantum field theory through such a limit. The numbers that admit the sharpest statements—mass and coupling

ladders, curvature ratios, shell thermodynamics, and rational witnesses in `HQIV_LEAN`—are objects of *discrete mathematics* on $\mathbb{N}$ (combinatorial growth, cumulative simplex ledgers) together with the ordered field $\mathbb{R}$ as used in the proof assistant for analysis, bounds, and phenomenological display. Smooth-manifold or continuum field notation, when it appears, is a *translation layer* for comparison with standard physics literature; it does not supersede the discrete definitions.

A continuum is not required, and in places would destroy these results. The discrete patch layer is *not* a numerical approximation to a more fundamental continuum theory awaiting future construction; it is the ontology. Where the literature treats “the continuum limit” as the goal, HQIV treats it as a *calculation approximation* for comparison with standard formulas. The continuum form is an optional convenience for comparison; the proved discrete statement is what carries the physics. In several load-bearing places—the curvature imprint $\delta_E(m)$ at shell m , the harmonic ladder masses $m_\tau \rightarrow m_\mu \rightarrow m_e$, the lock-in vev $v = \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$, the proton anchor at $m = \text{referenceM} = 4$, the baryon-asymmetry $\eta \propto 6^7 \sqrt{3}/(m+1)$, and the rational coefficients $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$ —taking a literal $m \rightarrow \infty$ or sub-Planck continuum limit would *erase* the discrete index that is doing the predictive work. Continuum forms of these objects are therefore not preferred refinements; they are degenerate, information-losing readouts of the same patch data.

An exception: $\mathfrak{so}(8)$ as a de-facto continuum algebra from a discrete construction. Principled exception to the discrete-only stance: the closed Lie algebra $\mathfrak{so}(8) \supseteq G_2 \cup \{\Delta\}$ generated from the Fano octonion left-multiplication table and the phase-lift generator $\Delta \in (e_1, e_7)$. Although $\mathfrak{so}(8)$ is a continuous (28-dimensional real) Lie algebra, it is built here entirely from *discrete* ingredients (the seven imaginary octonion units, a finite Fano table, and one $\pi/2$ rotation), and its closure to 28 dimensions is a finite linear-algebra certificate (see the `Hqiv.S08Closure` / `Hqiv.S08ClosureInterface` stack). The Lie-group structure is therefore a *de-facto continuum algebra obtained from a discrete construction*: continuous in parameter space, but with no continuum spacetime, no continuum measure, and no UV limit attached. When this manuscript or its companions write $\exp(t\Delta) \in \text{SO}(8)$ or treat $\mathfrak{so}(8)$ rotations as smooth, those are statements internal to a finite-dimensional Lie group whose generators are certified by discrete data; they are *not* a continuum-spacetime commitment and do not require a continuum-QFT scaffolding.

1. SCOPE

The Standard-Model Lagrangian bridge [1] constructs every sector of $\mathcal{L}_{\text{SM}}$ as a projection of the HQIV discrete octonion action. The strong gauge kinetic term $\mathcal{L}_G$ is obtained by restricting the abelian field-strength sum L_O^{kin} to the four octonion channels assigned to the strong sector by the algebraically fixed force-sector map. No additional vector fields, no non-abelian structure constants in the kinetic density, and no separate gluon self-interaction term appear in the proved bridge. The optional $\text{SU}(3)$ chart certificate supplies only a finite representation of the structure constants; it is explicitly excluded from the default library build and carries no dynamical content.

Simultaneously, the nucleon-binding note [3] realises all hadron and nuclear binding energies as a single 8×8 composite-trace functional on the same carrier:

$$E_{\text{bind}}(m) = \sum_{k=1}^{28} w_k \cdot (\text{latticeSimplexCount}(m) \cdot \alpha_{\text{eff}}(m)),$$

where the shell coupling $\alpha_{\text{eff}}(m)$ is the reciprocal of the O-Maxwell logarithmic running evaluated at the auxiliary field of shell m . The curvature-imprint ratio $\alpha = 3/5$ that fixes the running is the same combinatorial ratio forced by simplex counting on the null lattice.

The dynamic-mass-spectrum synthesis [2] supplies the continuous- ξ lift: the same curvature primitive

$$K(\xi) = \log \xi + \frac{\alpha}{2} (\log \xi)^2$$

normalised to the lock-in value as $\omega K(\xi)$ modulates the inner trapped-Casimir factor on the T12 Hopf shells. That factor is the direct source of the heavy lepton gap and, by the explicit Lean labelling, of the “heavy/gluonic” stabilisation.

The identification is therefore forced by the shared objects, not postulated.

2. THE T12 AND T13 WITNESSES

To make the argument self-contained, we briefly recall the geometric objects that supply the inner and outer Casimir factors. (Full technical details appear in the dynamic-mass-spectrum synthesis paper; here we give only the minimal physical and algebraic content required for the present claim.)

A *Hopf shell* in the discrete setting is an integrable three-dimensional complex on the octonion carrier, equipped with a curvature imprint α_n and a torsion matrix derived from the contact Beltrami operator on that shell. These are the finite, machine-checkable counterparts of the nested Hopf fibrations in Nielsen’s topological construction.

The *T12 witness* (the structure ‘exampleNonFactorableWitnessForIntegrableHopfShells’ in the Lean library) packages exactly three such shells, with winding numbers $n = 1, 2, 3$. Each shell carries its own torsion matrix obtained from the T11 phase-lift construction. The defining property of the witness is *non-factorability*: the three torsion matrices cannot be simultaneously block-diagonalised on any proper subspace of the eight-dimensional carrier. The structure is irreducible; the full octonion carrier participates as a single unit.

The *inner contact surfaces* of these three T12 shells are the locus of the trapped Casimir effect. The geometric factor generated on these surfaces (the trapping selection functional, modulated by the curvature primitive) is what the framework identifies as the source of binding energy and of the heavy stabilisation gap. In the Lean implementation this factor is explicitly labelled as supplying the “trapped, binding, heavy/gluonic” contribution.

The *T13 witness* (‘outerShellNeutrinoFluctuationWitness’) is the corresponding datum on the first shell beyond the reference lock-in surface. It counts discrete fluctuation modes (modeCount = 140) on the neutral singlet extension of the carrier — the right-handed neutrino channel realised by the 8_c conjugate representation together with a trivial summand. Coarse-graining these neutral outer fluctuations produces the characteristic suppression factor $1/140$. This is the outer Casimir contribution.

The effective mass scale at any horizon coordinate ξ is then the instantaneous ratio

$$\text{effective_casimir_scale_at_xi}(\xi) = \frac{\text{inner trapping from T12 surfaces, run by } \omega K(\xi)}{\text{outer suppression from T13 neutral fluctuations}}.$$

Everything that follows — the dynamic lepton spectrum, the temperature dependence, and the binding phenomenology — is generated from this single geometric asymmetry on one carrier.

3. THE STRONG SECTOR CARRIES NO NEW FIELDS

The force-sector assignment

$$\text{forceSector}(a) = \begin{cases} \text{EM} & a = 0, \\ \text{Weak} & a \in \{1, 2, 3\}, \\ \text{Strong} & a \in \{4, 5, 6, 7\} \end{cases}$$

is fixed once and for all by the octonion algebra and the colour-preferred axis of the $\mathfrak{so}(8)$ closure. The strong kinetic aggregate is

$$\mathcal{L}_G(A) = L_{O,\text{Strong}}^{\text{kin}}(A) = -\frac{1}{4} \sum_{a=4}^7 \sum_{\mu\nu} \frac{(F_{\mu\nu}^a(A))^2}{2}.$$

This is identical in form to the electromagnetic and weak projections; only the channel mask differs. The nuclear effective potential constructed in the Forces module realises the “strong” contribution as the linear term in the auxiliary field ϕ plus the curvature-imprint term $\delta_E(m) \times$ phase-fibre curvature on the octonion configuration. No new propagator, no new vertex, no new gauge boson.

What continuum QCD calls a gluon is, in this geometry, the effective description of the energetic cost of occupying or deforming the strong-channel components of the carrier inside the inner-Casimir trapping region of the T12 shells (the three irreducible integrable Hopf shells whose contact surfaces generate the trapped binding factor). The binding energy subtracted from constituent sums to obtain the physical nucleon mass is the same geometric deficit that appears, at the lepton level, as the heavy stabilisation gap.

4. UNIFORM NETWORK BINDING ACROSS ALL SCALES

The decisive structural fact, recorded in the bound-states module, is that binding is treated uniformly at every hierarchical level by a single 8×8 network formula on the $\mathfrak{so}(8)$ generators.

Binding at shell m takes the universal form

$$E_{\text{bind}}(m) = \sum_{k=1}^{28} w_k \cdot \alpha_{\text{eff}}(m),$$

where the w_k are state-dependent weights (matrix elements or traces over the 28 generators) and the coupling $\alpha_{\text{eff}}(m)$ is the effective fine-structure constant obtained from the O-Maxwell logarithmic running at the auxiliary field value of that shell. The same $\alpha_{\text{eff}}(m)$ — driven by the curvature-imprint ratio $\alpha = 3/5$ — appears for every layer.

The bound-states module makes the uniformity explicit by defining thin wrappers around one core routine: - `E_bind_QCD_from_network` (quarks to hadron) uses weights drawn from the colour/gluon sector of the 8×8 embedding; - `E_bind_nuclear_from_network` (nucleons to nucleus) uses weights from the strong residual plus electromagnetic contributions; - `E_bind_atomic` follows the same α_{eff} scaling inside the Schrödinger layer.

In other words, what is labelled “QCD binding” or “gluonic binding” is not a separate dynamical mechanism with its own force law. It is the identical composite-trace sum, with the weights w_k simply chosen to emphasise the four strong octonion channels ($a \in \{4, 5, 6, 7\}$) that the force-sector map assigns to the strong sector. The curvature running that produces $\alpha_{\text{eff}}(m)$ is the same running that appears in the inner-Casimir trapping factor.

5. BINDING ENERGY AND THE CURVATURE PRIMITIVE HAVE IDENTICAL FORM AND SCALE

All three layers—hadronic (“QCD”) binding, nuclear binding, and the heavy lepton gap—therefore evaluate instances of one and the same functional family:

- Discrete sampling: $\alpha_{\text{eff}}(m)$ obtained from $1/\alpha_{\text{GUT}} \times (1 + c\alpha \log(\phi(m) + 1))$ with $\alpha = 3/5$.
- Continuous lift: the identical running evaluated at $\phi(\xi) = 2\xi$, then modulated by the normalised curvature primitive $\omega K(\xi) = K(\xi)/K(5)$.
- Inner trapping: the selection functional $1 + c\alpha \log(1 + \frac{4}{3}\alpha)$ (phase-lift coefficient of the heavy T12 shell) multiplied by $\omega K(\xi)$ and divided by the outer suppression $1/140$.

The composite-trace weights that realise the shared nucleon binding at the lock-in shell are matrix elements on the same 28 generators. The subset of those generators lying in the strong channels supplies the dominant contribution to the binding that continuum language attributes to gluons. The small fractional shifts (for example the ~ 0.65 MeV electromagnetic block correction on the 939 MeV meta-harmonic anchor) are set by precisely the same $\alpha \log \phi$ curvature corrections that generate the few-percent deviations of the inner-Casimir scale from its lock-in value. Both the binding deficits and the heavy-gap dynamics are monotonic functions of the cumulative curvature integral along the temperature (or shell) ladder, governed by the single lattice parameter $\alpha = 3/5$.

The numerical agreement is therefore not an accidental match between independent phenomenological fits. It is the same geometric object — the curvature-driven network coupling evaluated on different projections of the carrier — sampled once for the lepton spectrum and once for the strong-channel weights that produce hadronic binding.

6. IMPLICATIONS AND CLAIM BOUNDARIES

The framework therefore contains a single abelian gauge theory (the octonion-indexed O-Maxwell action), a single geometric binding mechanism (the uniform 8×8 composite-trace network of the bound-states module, realised geometrically as inner-Casimir trapping on the T12 Hopf contact surfaces), and a single curvature primitive that runs every scale with cosmic temperature. The three Standard-Model gauge factors arise as the natural algebra projections of that carrier; the “gluon” is the continuum shadow of the strong-channel projection of this binding geometry.

What is deliberately left unclaimed is a dynamical explanation of confinement as a phase transition in an independent non-abelian gauge theory. Confinement, in the present ontology, is the statement that the inner-Casimir trapping region for the strong octonion components has a finite depth set by $\omega K(\xi)$ and the T12 torsion data; free colour charges would require promoting a strong-channel excitation above that depth, which is geometrically disfavoured exactly as the right-handed neutrino channel is suppressed by the outer Casimir. A future non-abelian holonomy upgrade may enrich the kinematics, but it is not required for the binding or mass phenomenology already realised.

The last independent dimensionful input beyond the discrete lattice combinatorics and the choice of “now” slice on the temperature ladder has therefore been removed. The strong sector, like the electroweak and gravitational sectors, is geometry all the way down.

APPENDIX A. LEAN MODULE INDEX

Force-sector map.: `Hqiv/Physics/Forces.lean`, definition `O_component_to_sector`; inductive `ForceSector.Strong`.

Strong kinetic projection.: `hqiv-lean/Hqiv/Physics/Forces.lean` (sector map) and the SM Lagrangian bridge module; theorem `L0-kinetic-split` (three-sector split of the abelian octonion kinetic).

Dynamic inner-outer Casimir scale.: `Hqiv/Physics/HopfShellBeltramiMassBridge.lean`, definition `effective_casimir_scale_at_xi`; the comment “Inner Casimir (trapped, binding, heavy/gluonic)” appears in the surrounding block. Uses `omegaK_xi` from `Hqiv/Physics/ContinuousXiPath.lean` and `Hqiv/Physics/ContinuousXiCoupling.lean`.

Curvature primitive.: `Hqiv/Physics/ContinuousXiCoupling.lean`, `continuousCurvaturePrimitive` and `omegaKContinuous`. Discrete sampling via `one_over_alpha_eff` in `Hqiv/Physics/SM\GR\_Unification.lean`.

Uniform 8×8 network binding.: `hqiv-lean/Hqiv/Physics/BoundStates.lean`, the single core definition `E_bind_from_network` together with the thin wrappers `E_bind_QCD_from_network` and `E_bind_nuclear_from_network`. All hierarchical binding (hadronic through atomic) uses the identical curvature-driven coupling $\alpha_{\text{eff}}(m)$ on the 28 $\mathfrak{so}(8)$ generators; only the weights w_k change with sector. Re-export of `alphaEffAtShell`. Concrete witness values in `hqiv-lean/Hqiv/Physics/QuarkMetaResonance.lean` and `hqiv-lean/Hqiv/Physics/DerivedNucleonMass.lean`.

Trapping selection functional.: `Hqiv/Physics/HopfShellBeltramiMassBridge.lean`, `trappingSelectionFromHeavyHopfShellWithAlpha` (phase-lift from T12 heavy shell).

Lattice curvature imprint.: `Hqiv/Geometry/OctonionicLightCone.lean` (or `AlphaGammaForcedByLattice`), theorems forcing $\alpha = 3/5$, $\gamma = 2/5$ from simplex counting.

REFERENCES

- [1] Steven Jr Ettinger. Building the standard model lagrangian from the HQIV discrete action: A sector-by-sector lean bridge. Preprint, HQIV-LEAN repository ([papers/sm_lagrangian/](#)), 2026. In preparation. To be deposited at Zenodo (HQIV community) on Lean PR build completion; this entry will be updated with the minted DOI on publication.
- [2] Steven Jr Ettinger. Electroweak scale and fermion masses from horizon readouts: First-principles electroweak closure inside the HQIV lean library. Preprint, HQIV-LEAN repository ([papers/lean_to_mass_spectrum/](#)), 2026. In preparation. To be deposited at Zenodo (HQIV community) on Lean PR build completion; this entry will be updated with the minted DOI on publication.
- [3] Steven Jr Ettinger. Nucleon binding from HQIV composite-trace weights: Lock-in masses, lapse readout, and neutron stability scaffold. Preprint, HQIV-LEAN repository ([papers/nucleon_binding/](#)), 2026. In preparation. Tier-2 #7: 8×8 network binding at `referenceM`; companion to baryogenesis and precursor to BBN.

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